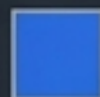


Mastering Retrieval-Augmented Generation (RAG)

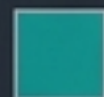
The Engineering Architecture of Truthful AI

An End-to-End Blueprint for Pipeline Design, Evaluation,
and Advanced Implementations.

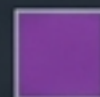
System Colour Architecture



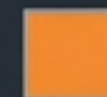
Data & Storage



Retrieval & Search



AI Models & Generation



Evaluation Metrics

What is our company
leave policy?

HALLUCINATIONS

The model invents
plausible but false facts
when it lacks specific
training data.

OUTDATED KNOWLEDGE

The model's worldview is
frozen at its last
training cut-off date.



Fails to answer
from static
memory.

LIMITED CONTEXT

Constrained context
windows lead to memory
dilution and expensive
inference.

NO PRIVATE ACCESS

Cannot read internal,
proprietary, secure, or
enterprise-specific
documents.

Takeaway: Standalone models generate from static memory.
We need architectures that generate from dynamic truth.

RAG vs CAG: What's the Difference?

RAG (Retrieval-Augmented Generation)

Search the world before you answer.

- **Data Source:** External docs, databases, web files.
- **Mechanism:** Uses Vector Databases and search engines.
- **Characteristics:** Slower (retrieval adds latency), High hallucination reduction.

CAG (Context-Augmented Generation)

Remember the conversation before you answer.

- **Data Source:** Conversation history, user profiles, session memory.
- **Mechanism:** No Vector DB; uses existing context memory.
- **Characteristics:** Faster (no retrieval step), Moderate hallucination reduction.

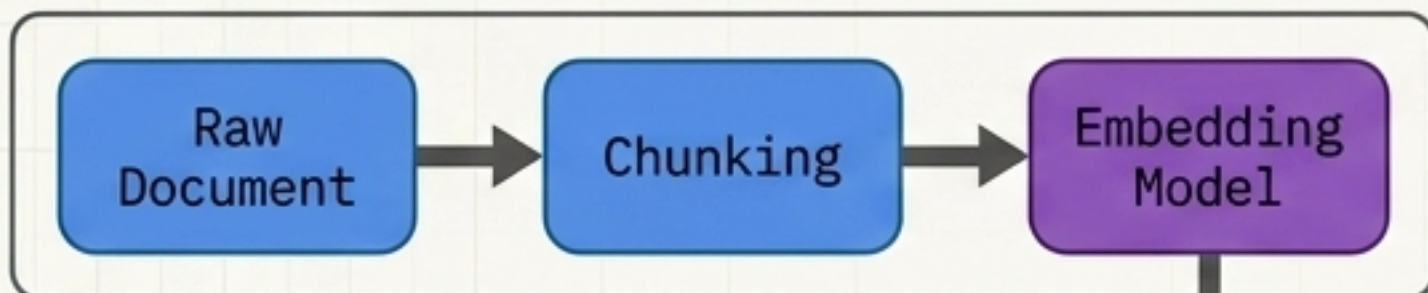


Best Used Together.

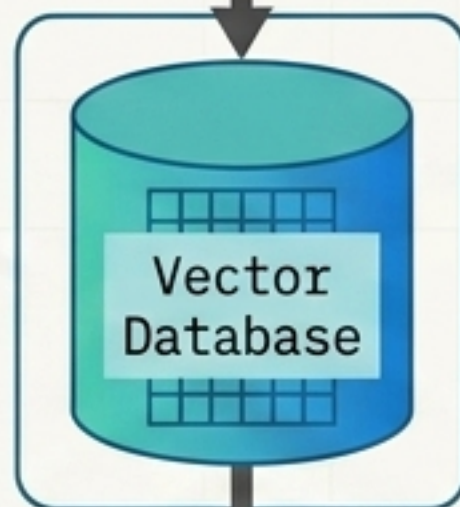
RAG provides external knowledge;
CAG provides internal memory.

End-to-End RAG Pipeline: The Master Blueprint

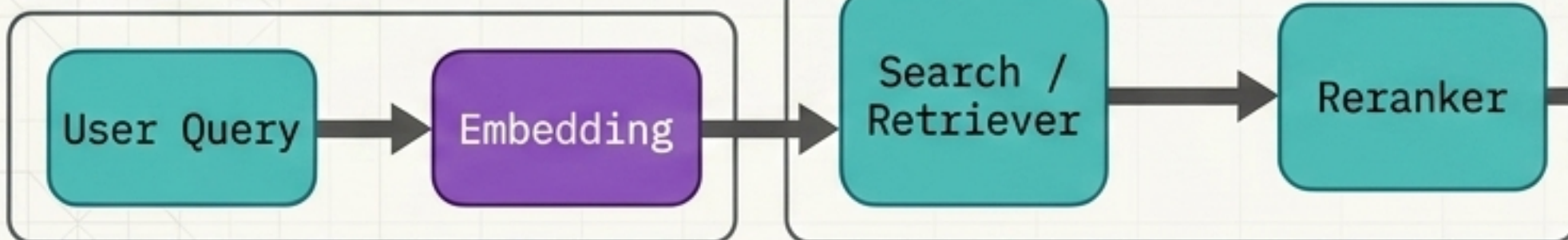
Stage 1: Ingestion



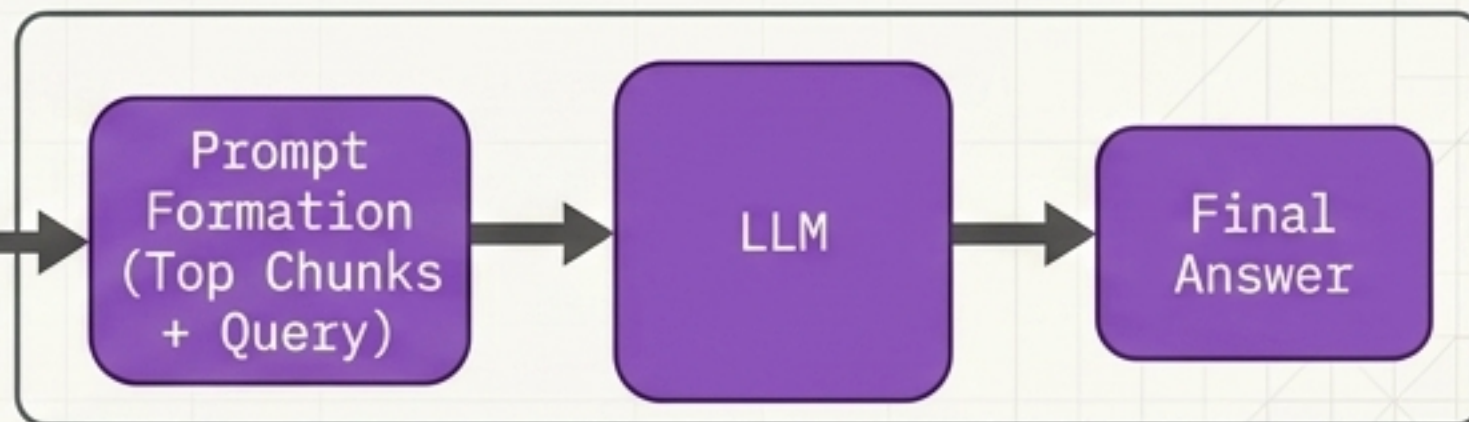
Stage 2: Storage



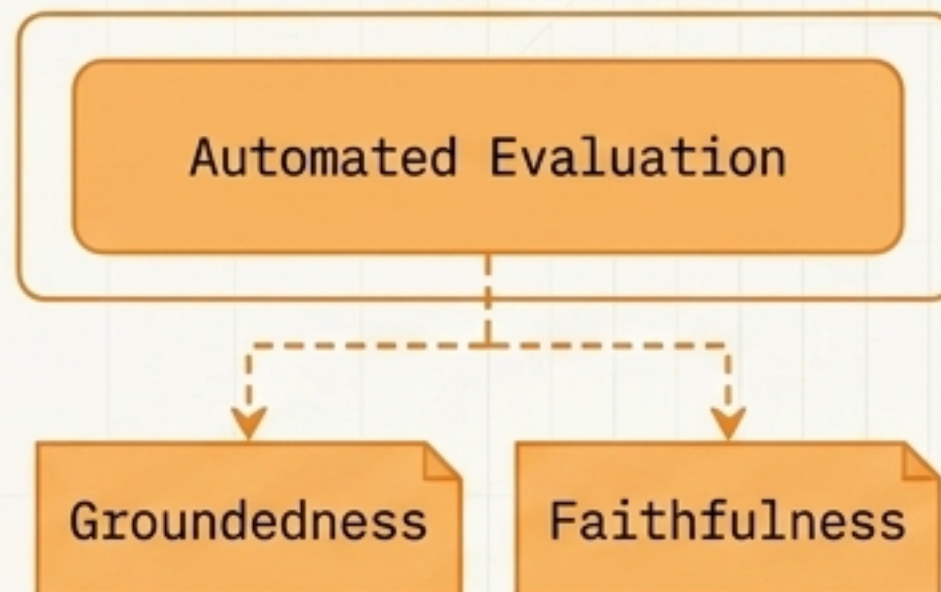
Stage 3: Retrieval



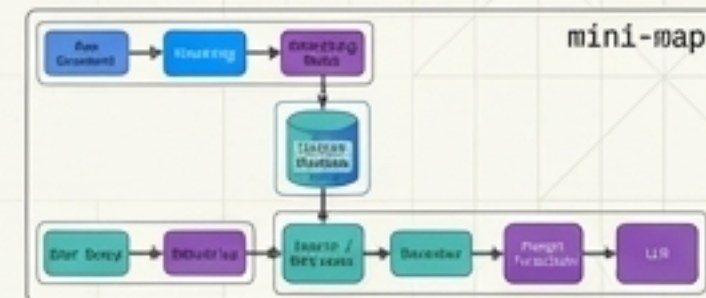
Stage 4: Generation



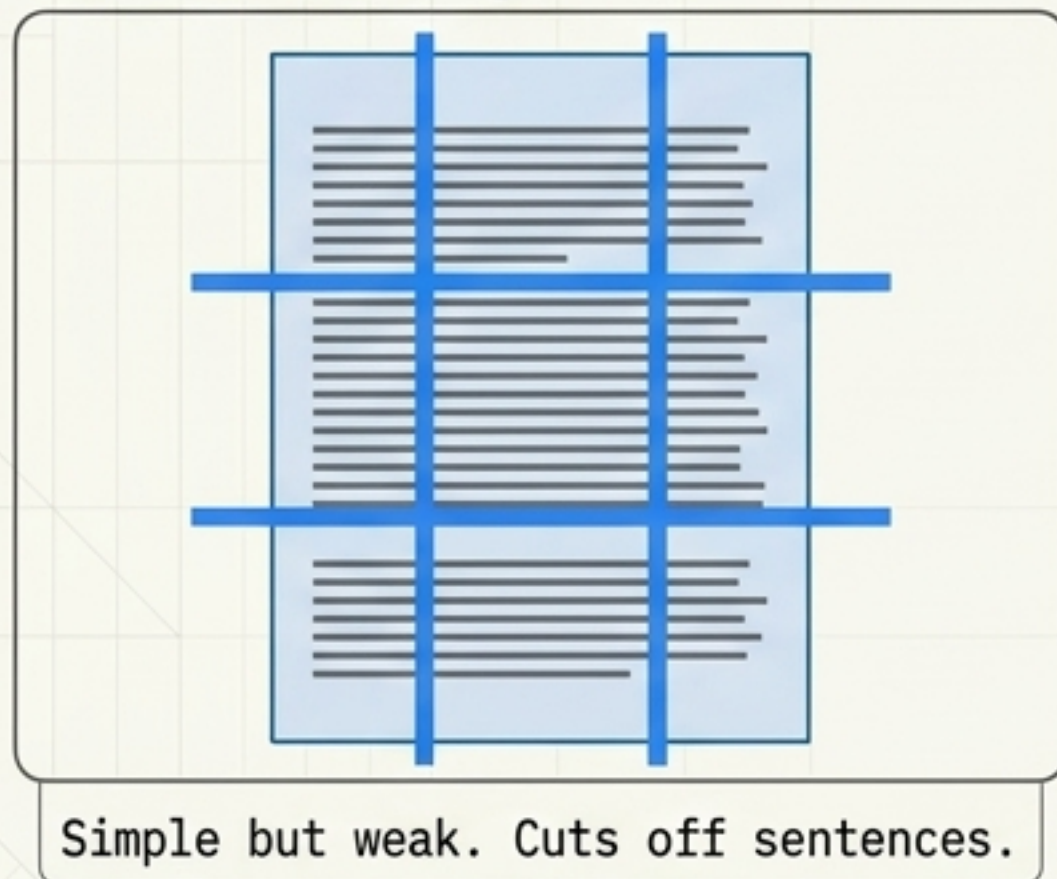
Stage 5: Evaluation



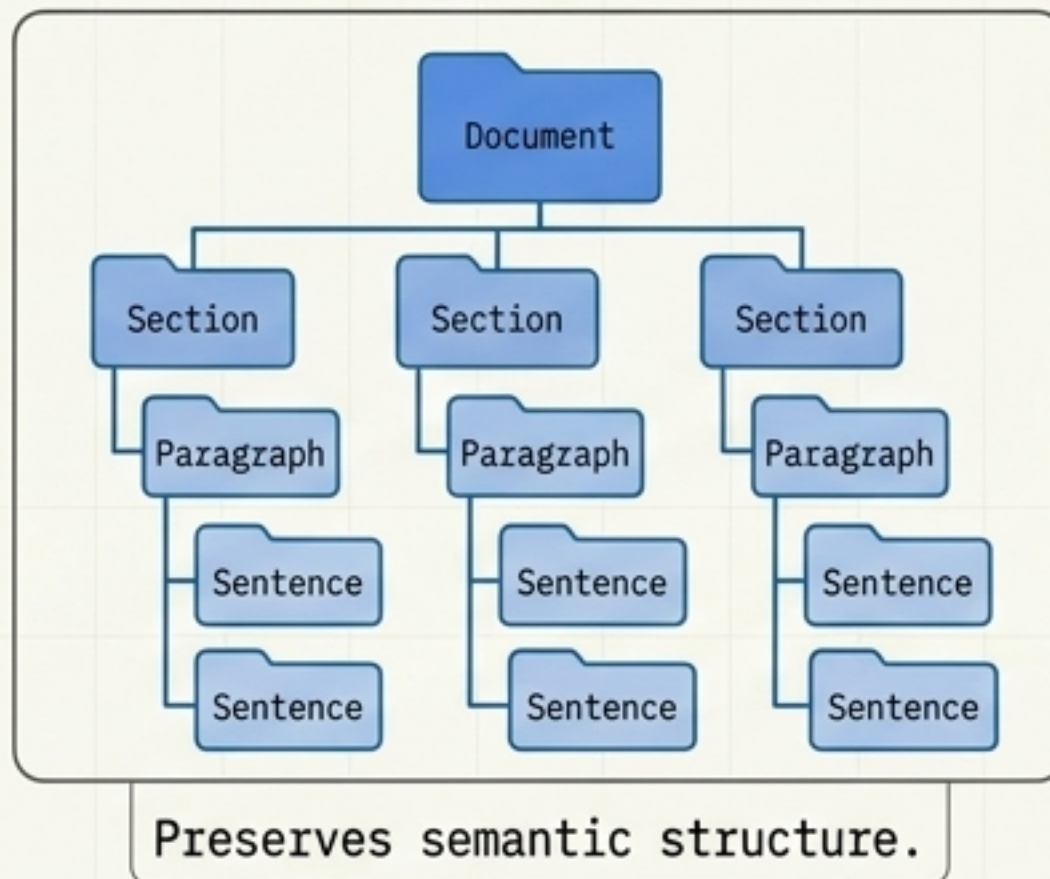
Phase 1: Ingestion & Document Splitting



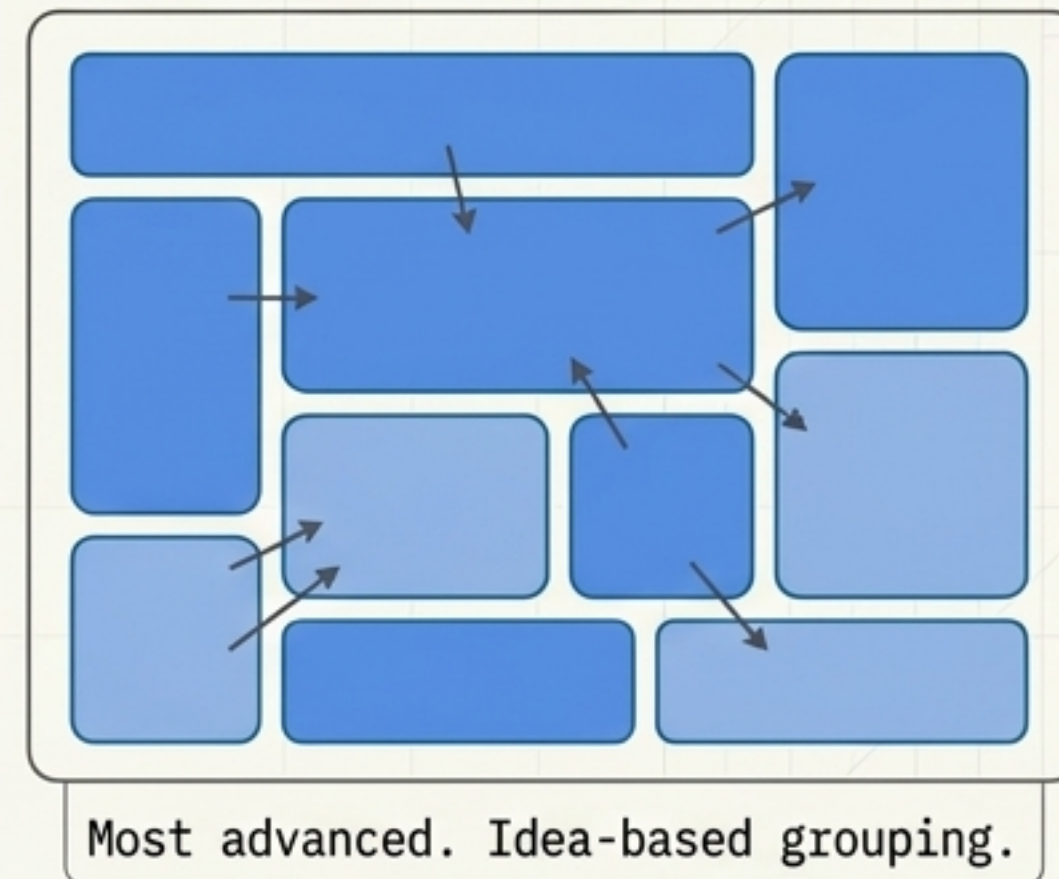
Fixed Chunking



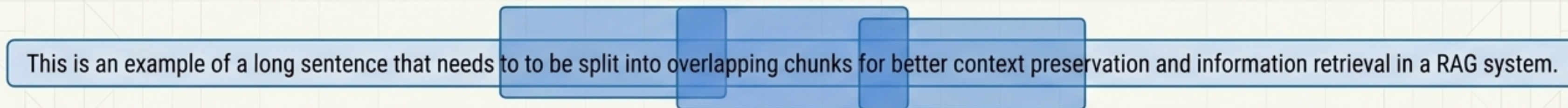
Recursive Chunking



Semantic Chunking



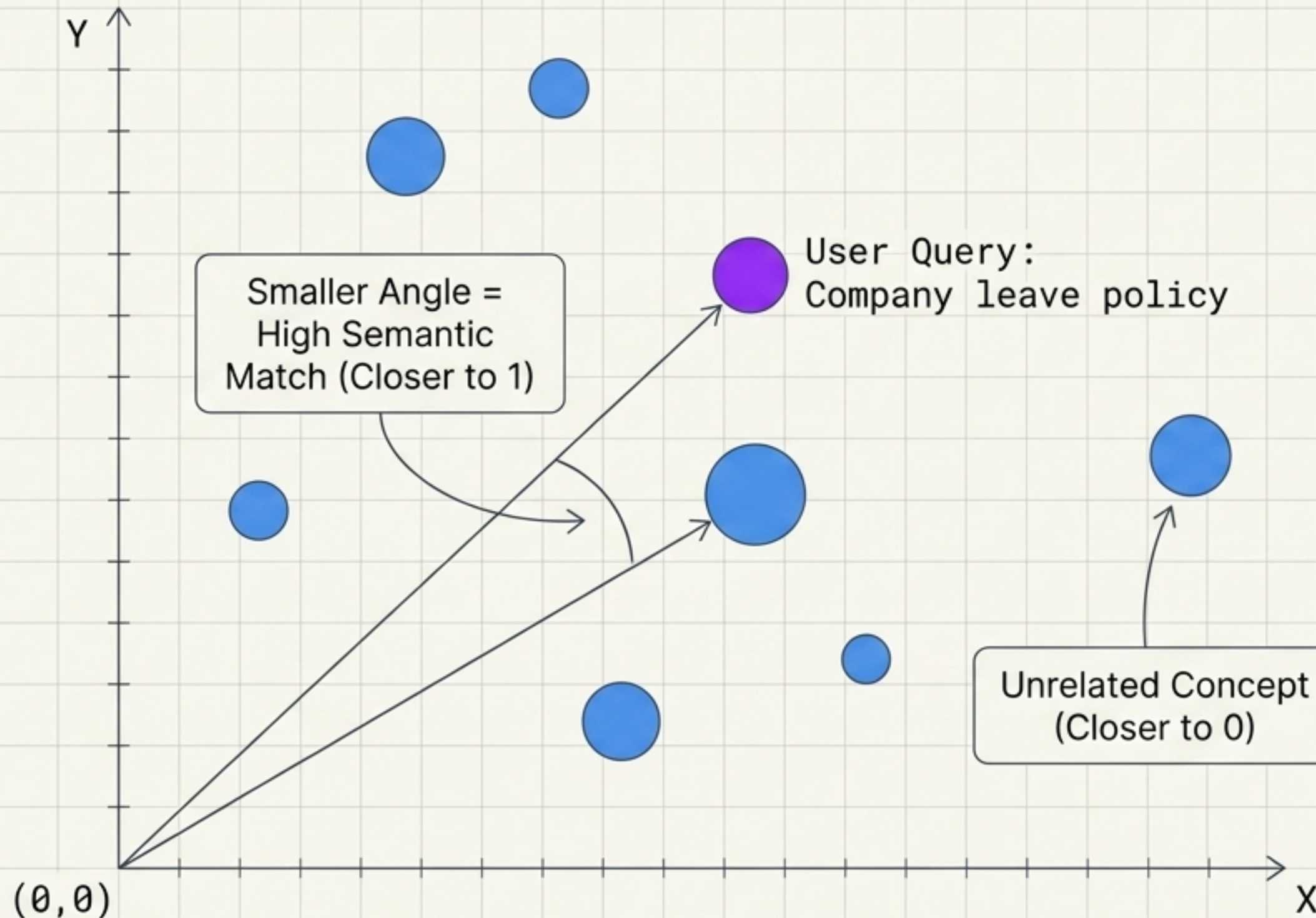
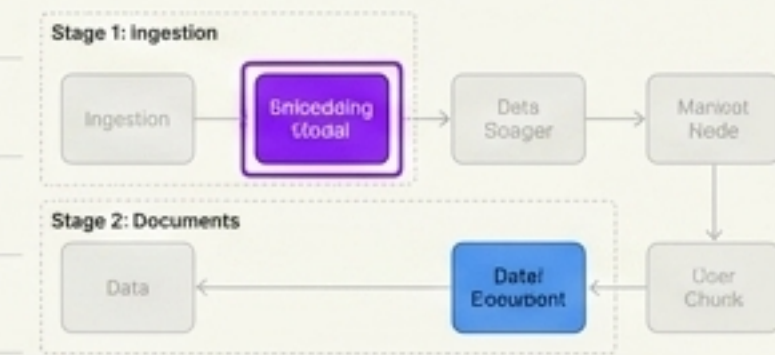
The Overlap Solution: Sliding Window Chunking



Overlapping chunks at the seams prevents context loss and missing information.

⚠ Bad chunking causes poor retrieval, missing information, and subsequent hallucinations.

Phase 1: Translation to Cosine Similarity Space



$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{|A||B|}$$

Industry Standard Embedding Models:

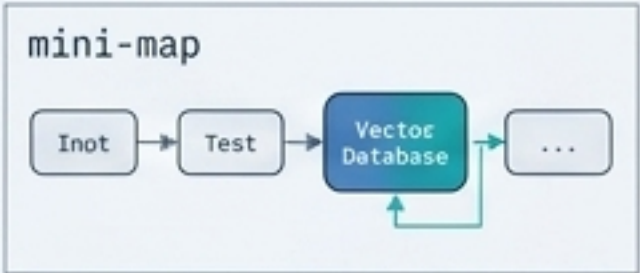
OpenAI embeddings

Sentence Transformers

BGE

E5

Phase 1: High-Dimensional Storage



Traditional Relational DB

Query: "vacation"  Fails: Relies on exact keyword matching.

				holiday		

Vector Database

Query: "vacation"  Succeeds: Captures underlying semantic similarity.

holiday

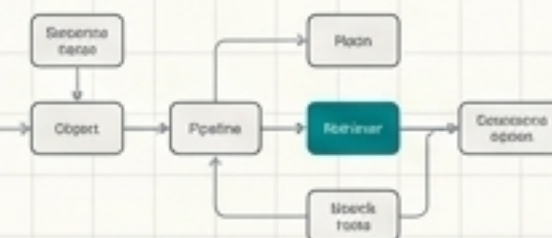
leave

time off

Popular Vector Infrastructure

Pinecone	Weaviate	ChromaDB	FAISS	Milvus
(Fully managed)	(Open-source)	(Lightweight local)	(Meta's scaling library)	(Massively scalable)

Phase 2: The Search Spectrum



Dense Retrieval

Mechanism: Uses **Embeddings**.

Strengths: Deep semantic understanding. Finds conceptual matches even if vocabulary differs.

Sparse Retrieval

Mechanism: Uses Keywords (e.g., BM25 algorithm, Elasticsearch).

Strengths: Exact phrasing, specific jargon, names, and acronym matching.

Hybrid Retrieval The Gold Standard

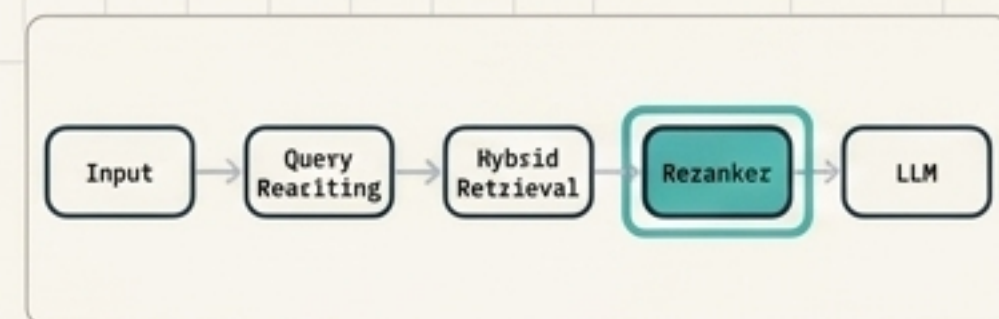
$$\left[\begin{array}{c} \text{Dense} \\ \text{Semantic} \\ \text{Search} \end{array} \right] + \left[\begin{array}{c} \text{Sparse} \\ \text{Keyword} \\ \text{Search} \end{array} \right] = \text{Hybrid Precision}$$

Dramatically improves retrieval quality by combining deep meaning with exact term relevance.

Pipeline Booster: **Query Rewriting**

Transforming a vague user query into a highly specific search string before retrieval to boost hit rates.

Phase 2: The Funnel of Precision



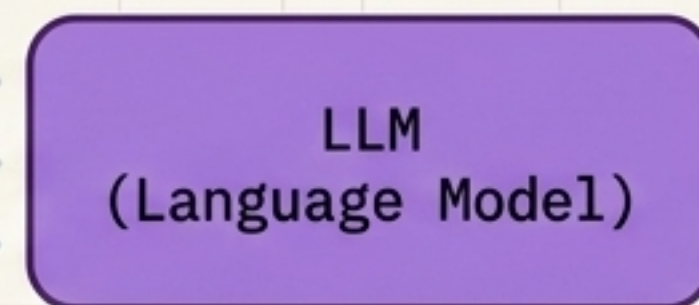
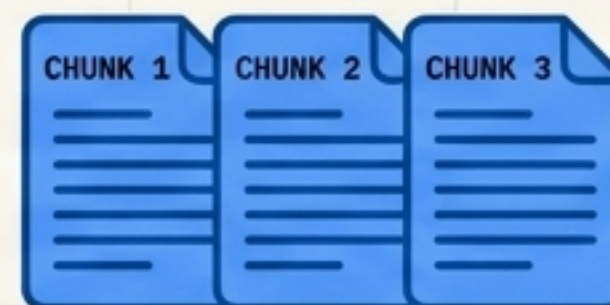
Initial Retrieval (Bi-Encoder / Vector DB)

Specs:

- Speed: Extremely Fast
- Accuracy: Approximate
- Scope: Broad search
- Output: Captures 100+ partially relevant chunks.

Cross-Encoder Reranking

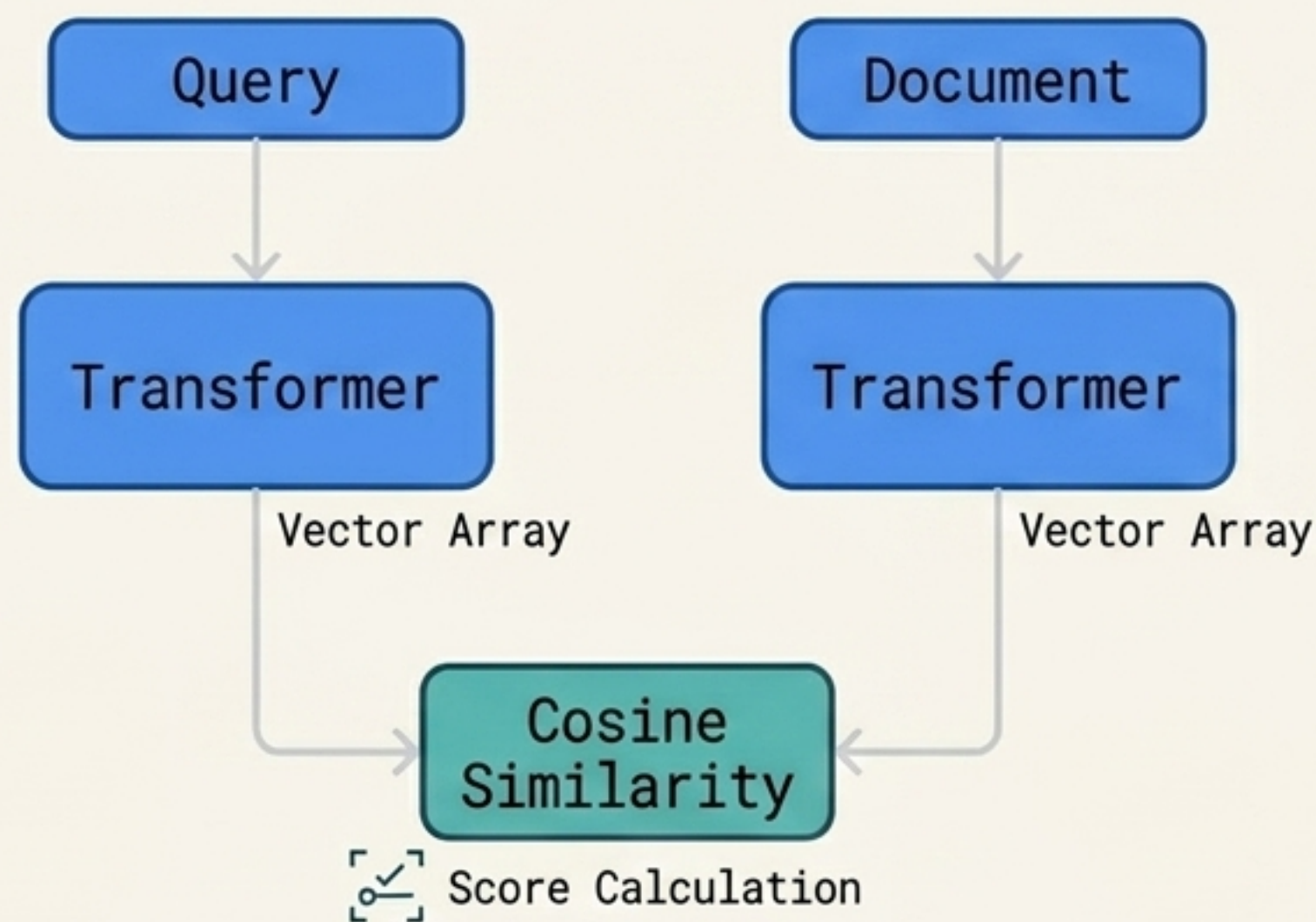
- Speed: Computationally Slower
- Accuracy: Highly Precise
- Scope: Fine-grained ranking
- Output: Reorders chunks using deep relevance analysis to output only the Top 3 chunks for the LLM.



Phase 2: Deep Dive on Encoder Architectures

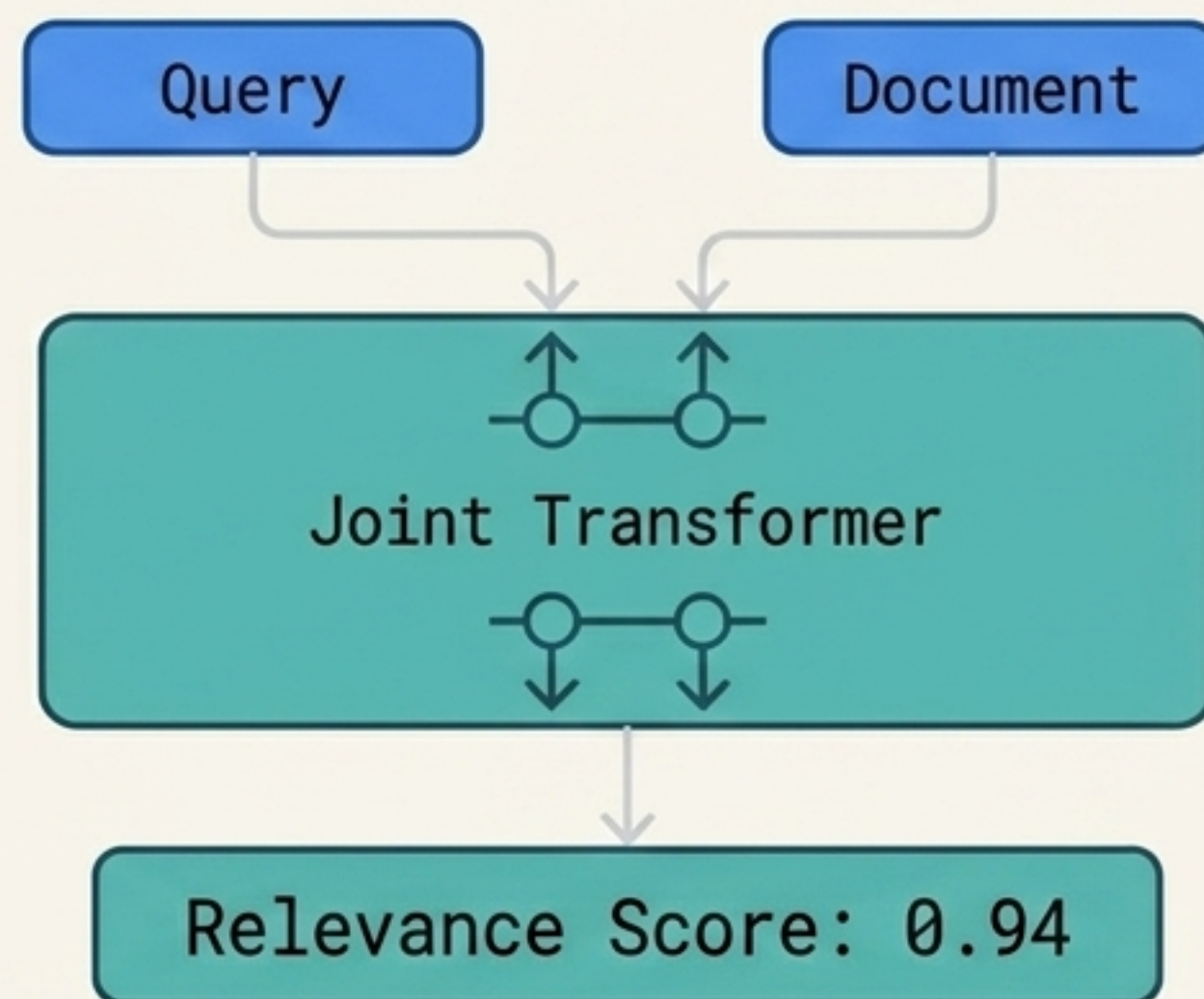


Bi-Encoder (Retrieval Focus)



Fast, Independent Processing,
Approximate Similarity.

Cross-Encoder (Reranking Focus)

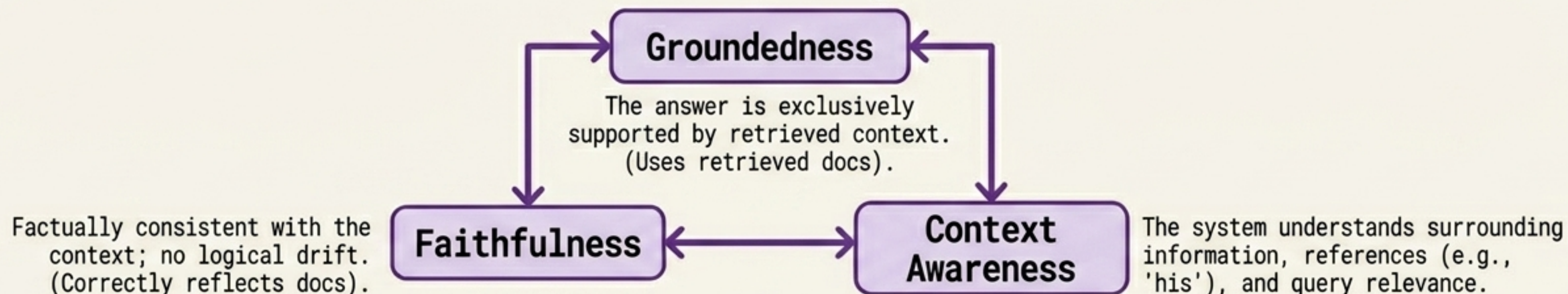


Computationally Expensive, Joint Processing,
Highly Accurate Interaction.

Phase 3: Generation & The Truth Triangle



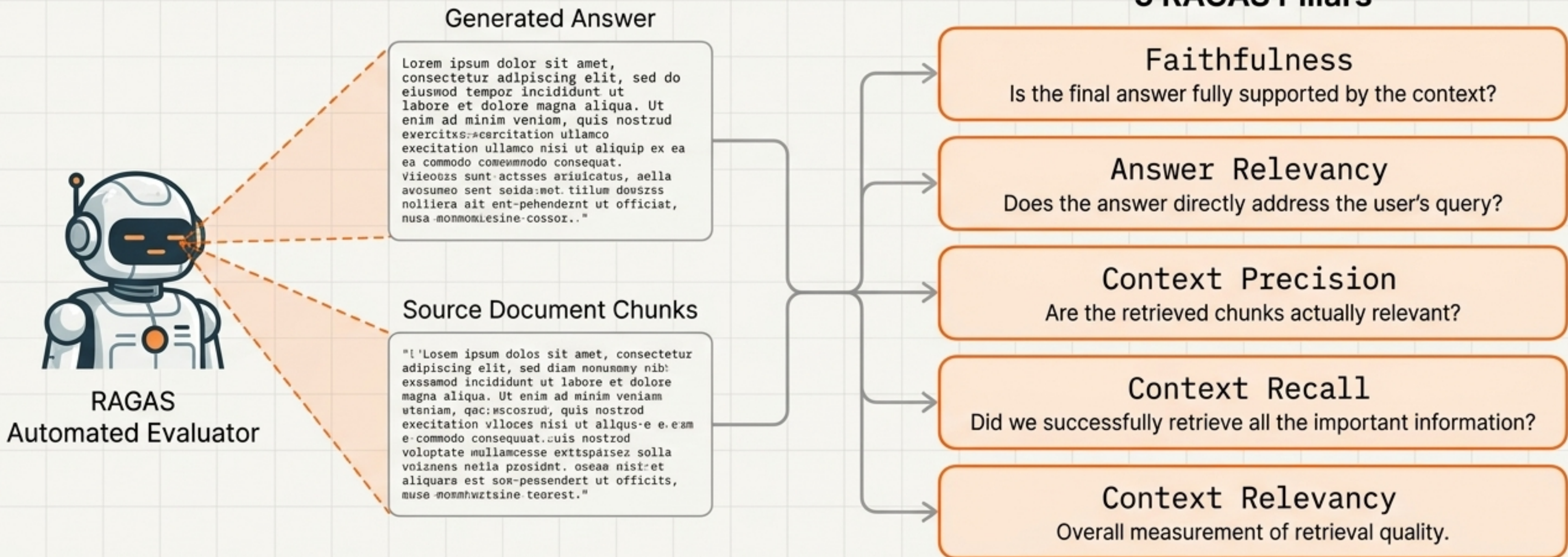
The Truth Triangle



Hallucination Taxonomy (Error vs. Fix)

Type	Error	Fix
Factual	Wrong facts generated.	Strict Grounding Prompts.
Citation	Fake sources invented.	Strict Context Limits.
Logical	Incorrect reasoning from true facts.	Cross-Encoder Reranking & superior LLMs.

Phase 3: Automated Evaluation (RAGAS)



RAGAS: Automated, label-free evaluation for scaling RAG pipelines.



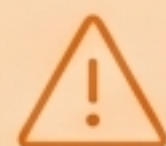
Phase 3: Legacy Lexical Evaluation

BLEU Score

- **Focus:** Precision (How much of the generated text appears in the reference?)
- **Mechanism:** n-gram overlap.
- **Primary Use Case:** Translation evaluation.

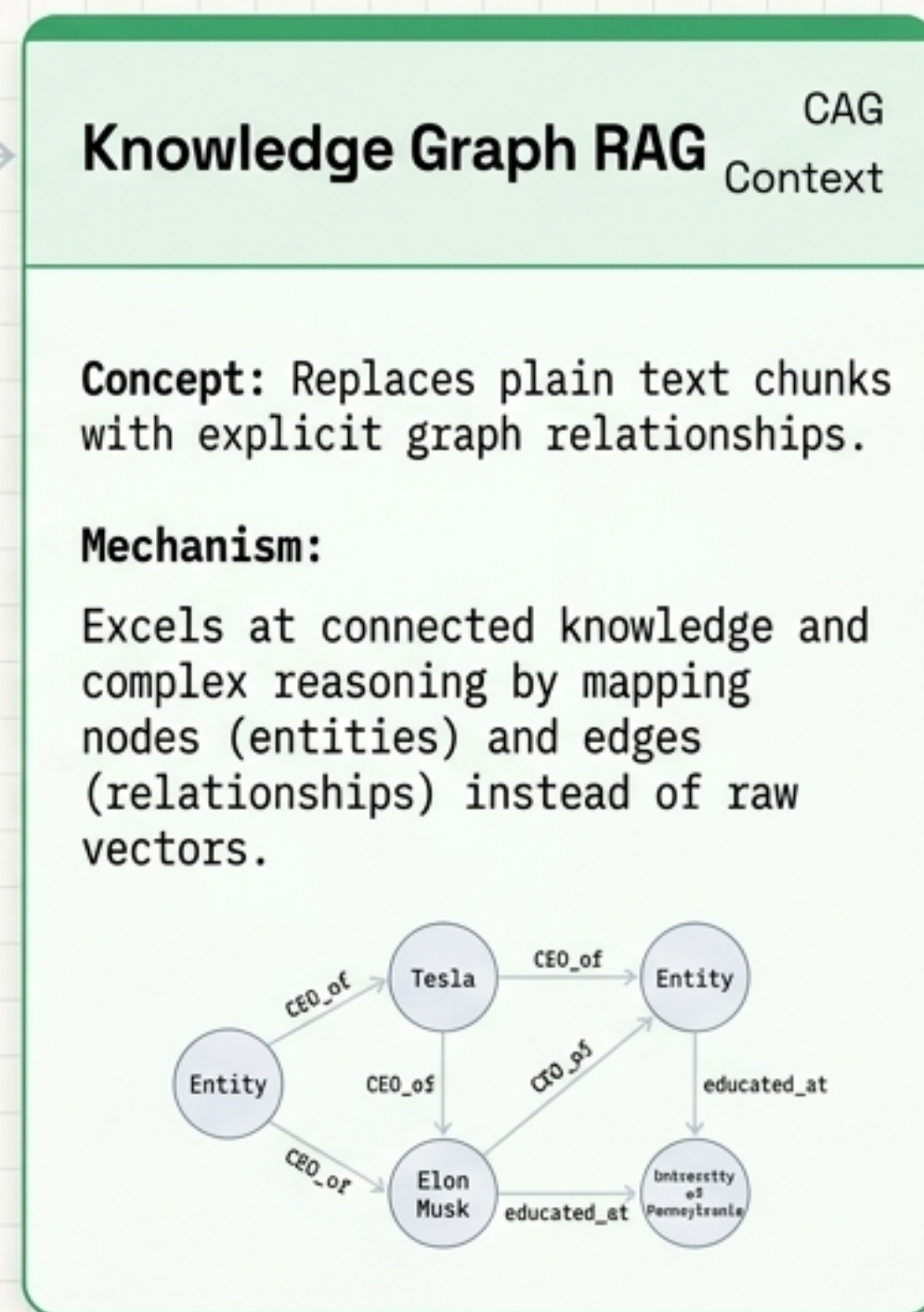
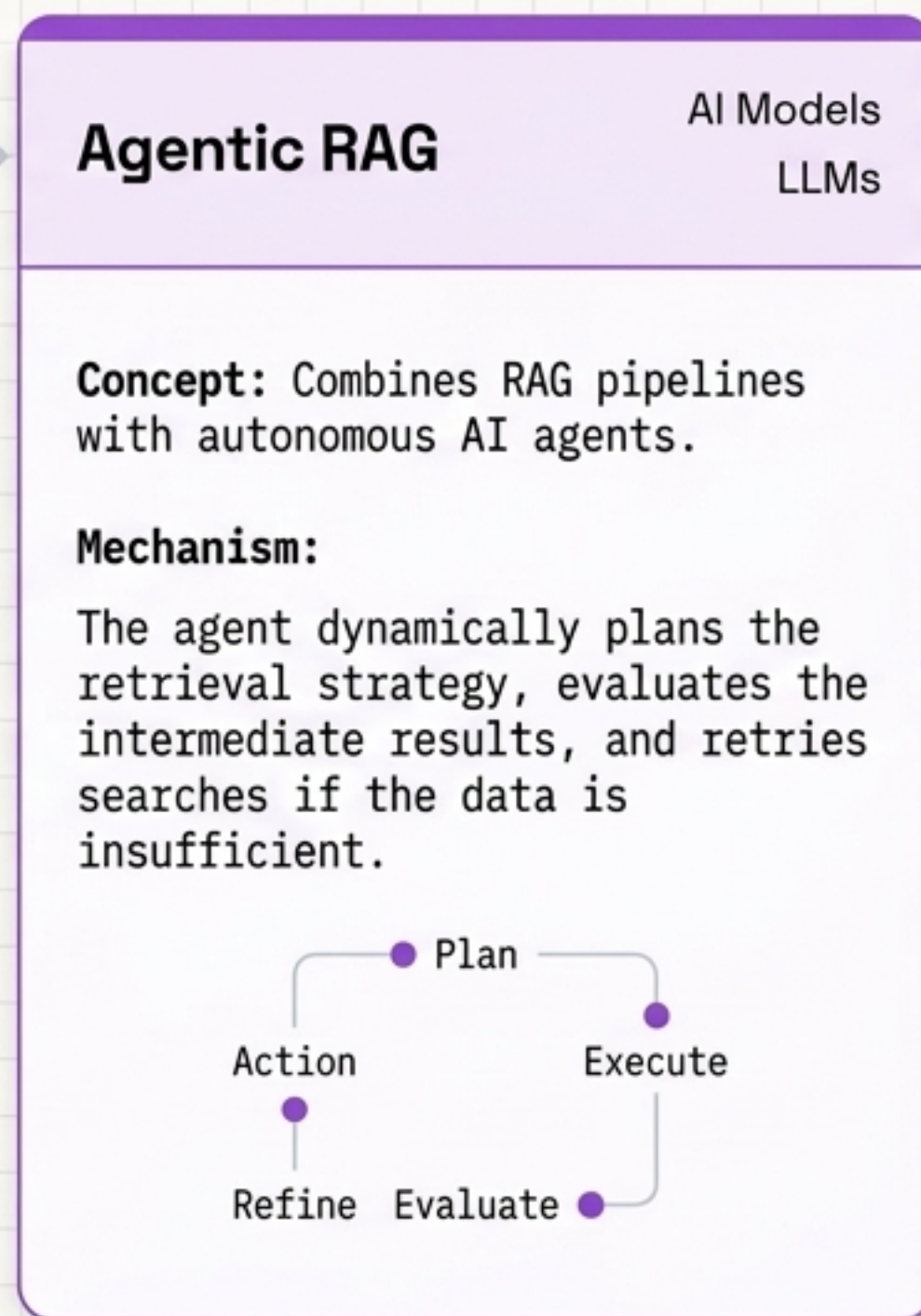
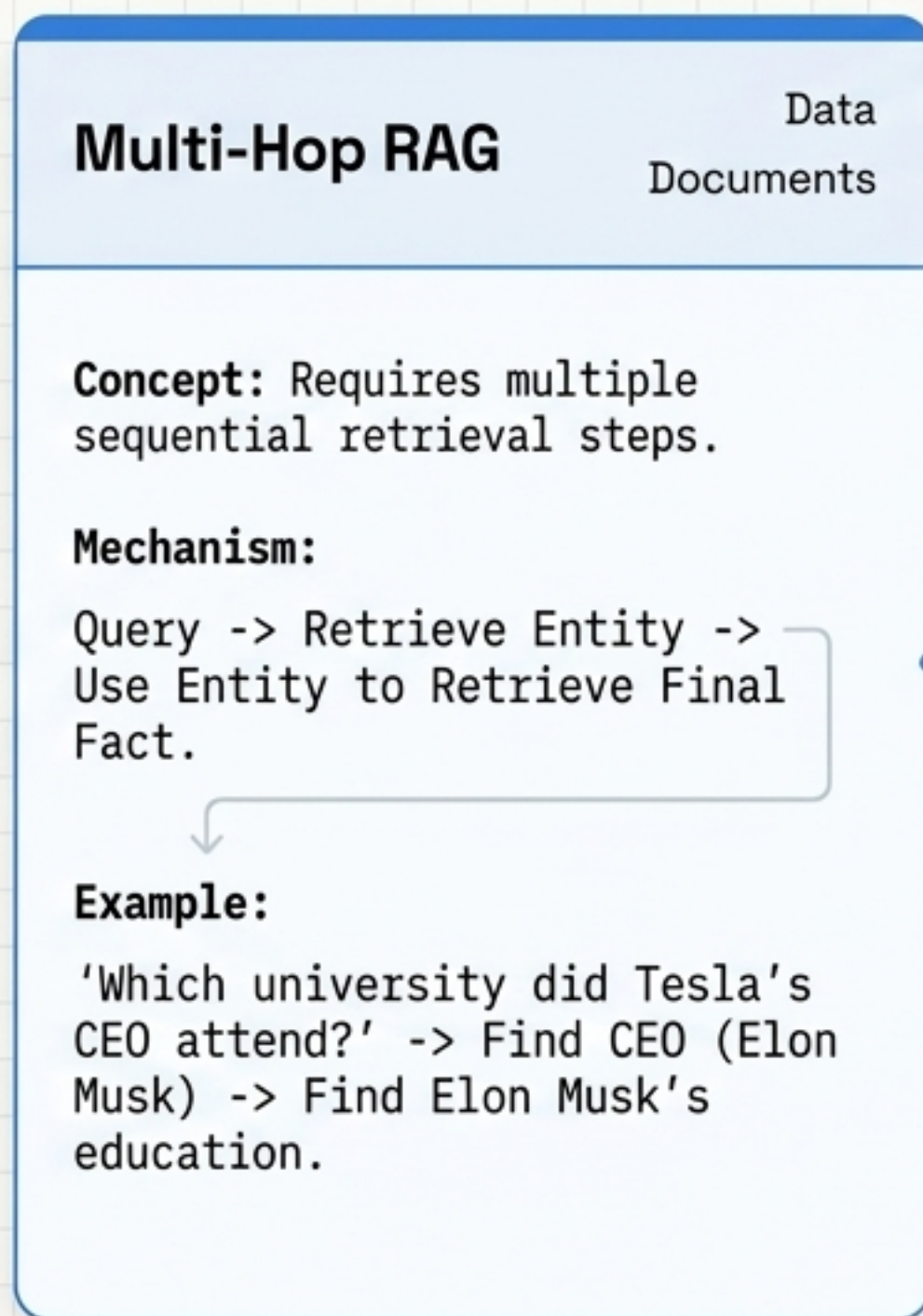
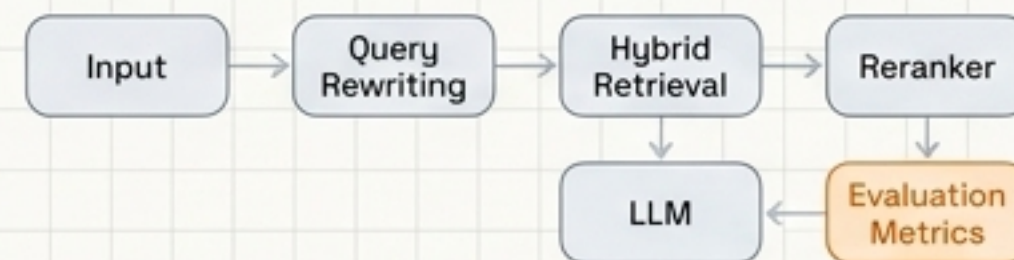
ROUGE Score

- **Focus:** Recall (How much of the reference text was successfully captured by the generated text?)
- **Mechanism:** Overlap types (ROUGE-1 unigram, ROUGE-2 bigram, ROUGE-L longest common subsequence).
- **Primary Use Case:** Summarisation evaluation.



Both are lexical metrics that risk failing on pure semantic understanding. They measure exact word overlap, not underlying meaning.

The Frontier: Advanced Architectures



The Engineer's Cheat Sheet (One-Line Memory Tricks)

Embeddings	= Meaning vectors in spatial dimensions.
Chunking	= Splitting documents to preserve context limits.
Retriever	= Finds relevant information via semantic/keyword search.
Reranker	= Improves ranking precision.
Cross Encoder	= Deep, joint comparison of query and document.
Grounding	= Answer strictly supported by retrieved evidence.
Faithfulness	= Factual truthfulness; no logical drift.
RAGAS	= Automated framework for RAG evaluation.
BLEU	= Precision-focused Translation metric.
ROUGE	= Recall-focused Summarisation metric.

Best Practices: Granular chunking, Hybrid Retrieval, Cross-Encoder Reranking, Strict Grounding Prompts.